Concentration of Measure

Proof notes, organized by engine

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dyed in indigo, madder & weld; ruled in iron-gall. woven on LOOM.

OVERVIEW · THE EIGHT ENGINES

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> In one line: a function of many *weakly dependent* variables is nearly constant. Every concentration inequality is one **engine** for proving this; they differ only in the structure they consume: independence, bounded differences, convexity, or curvature and isoperimetry. Learn the engine before the formula.

Trigger. You want a small-probability bound on $f(X_1, \dots, X_n)$ deviating far from its mean/median.

Ask yourself one question: *which structure do I have?* Then reach for the engine below.

Cheat-sheet: which engine to use

Engine	Structure it eats	Main tools / section
Cramér–Chernoff	independence + MGF	Laplace transform, optimize λ, rate function (§1)
Sub-Gaussian / sub-exp.	*shape* of the tail	ψ-Orlicz norms, Bernstein (§2)
Bounded diff. / martingale	change one coordinate	Azuma, McDiarmid (§3)
Variance / entropy	tensorization	Efron–Stein, Herbst, log-Sobolev (§4)
Geometry / isoperimetry	metric *and* measure	sphere/Gaussian isoperimetry, Borell–TIS (§5)
Talagrand convex dist.	convexity + product space	convex-distance inequality (§6)
Matrix concentration	spectrum + trace exponential	Lieb, matrix Bernstein (§7)
Suprema / chaining	metric entropy	Dudley, generic chaining (§8)

> Reading guide: §1–§2 are the trunk (the continuation of Appendix A); read them until fluent. §3–§4 push ``sums'' to ``any bounded-difference function''. §5–§6 are the geometric heart that gives ``*measure* concentration'' its name. §7–§8 are two exits toward ML/TCS.

CONTENTS

1 THE CRAMÉR–CHERNOFF ENGINE

□□□□□ The trunk. Every later engine just works on one step of this pipeline.

> In one line: to bound the probability that a random quantity strays far from its mean by something **exponentially small**, exponentiate the tail through the Laplace transform (the MGF) and pick a good λ to push the right-hand side down. The whole subject is variations on this single move.

The paradigm

Trigger. You want an exponentially small bound on $\Pr[X \geq a]$, with the MGF of X under control.

For every $\lambda > 0$,

$$\Pr[X \geq a] = \Pr\big[e^{\lambda X} \geq e^{\lambda a}\big] \leq \mathbb{E}[e^{\lambda X}]\, e^{-\lambda a} \qquad (\text{Markov on } e^{\lambda X}),$$

then take the infimum over λ. Writing the cumulant generating function $\psi_X(\lambda) = \log \mathbb{E}[e^{\lambda X}]$, ⇝ mgf

$$\log \Pr[X \geq a] \leq -\psi_X^*(a), \qquad \psi_X^*(a) = \sup_{\lambda > 0} \big(\lambda a - \psi_X(\lambda)\big),$$

where ψ_X^* is the **Legendre transform** (the rate function) of ψ_X.

> Three-line heuristic: (1) independence ⇒ MGFs *multiply*, which is the *only* reason a ``sum'' gets an exponential bound; (2) *center* first, $X \leftarrow X - \mathbb{E}X$, for the cleanest argument; (3) bound each $\mathbb{E}[e^{\lambda X_i}]$ by one *termwise convex* estimate. Every theorem in the subject just swaps that estimate and that λ.

Cheat-sheet: the three opening moves

Tool	Setting	Tail bound / when to use
Markov	$Y \geq 0$	$\Pr[Y > \alpha \mathbb{E}Y] < 1/\alpha$; opening move, first-order info only
Chebyshev	$\operatorname{Var} X < \infty$	$\Pr[\lvert X - \mathbb{E}X\rvert > t] \leq \operatorname{Var} X / t^2$; second-order info
Chernoff (master)	MGF exists	$\Pr[X \geq a] \leq e^{-\psi_X^*(a)}$; want an exponential tail

Main results & proof skeletons

> **Theorem 1.1** *(Chernoff master bound).* If the MGF of X is finite near 0, then $\Pr[X \geq a] \leq e^{-\psi_X^*(a)}$.

Proof. Skeleton: *(i)* Markov on $e^{\lambda X}$; *(ii)* take logs to get $\log \Pr[X \geq a] \leq \psi_X(\lambda) - \lambda a$; *(iii)* take the infimum over $\lambda > 0$. [fill in: add convexity of ψ_X, the attaining λ, and the equality case] ∎

> **Lemma 1.2** *(Additivity: independence makes ψ add).* $X_1, \dots, X_n$ independent ⇒ $\psi_{\sum_i X_i} = \sum_i \psi_{X_i}$.

Proof. [fill in: one line: $\mathbb{E}[e^{\lambda \sum X_i}] = \prod \mathbb{E}[e^{\lambda X_i}]$, take logs] ∎

Example 1.3 *(The sub-Gaussian seed: a symmetric ±1 sum).* Let ε_i be independent uniform ±1 and $S_n = \sum_{i\le n} \varepsilon_i$. From $\cosh\lambda \le e^{\lambda^2/2}$ we get $\psi_{S_n}(\lambda) \le \lambda^2 n/2$, and $\lambda = a/n$ gives

$$\Pr[S_n \ge a] \le e^{-a^2/2n}, \qquad \Pr[|S_n| \ge a] \le 2e^{-a^2/2n}.$$

[fill in: prove $\cosh\lambda \le e^{\lambda^2/2}$ by termwise Taylor comparison; this is the seed inequality of the whole subject]

> Why it goes this way: $\cosh\lambda \le e^{\lambda^2/2}$ is the *seed inequality*. Replace it by a different termwise estimate (Hoeffding's lemma → bounded variables, Bennett/Bernstein → variance-aware) and you grow §2–§4. Swap the estimate, swap λ, that is all.

Tightness: Chernoff gets the large-deviation rate right

Theorem 1.4 *(Cramér, matching lower bound).* Under mild conditions $\lim_{n\to\infty} \frac{1}{n}\log\Pr\left[\frac{1}{n}\sum_i X_i \ge a\right] = -\psi^*_{X_1}(a)$.

Proof. Skeleton: the upper bound is Thm 1.1+Lem 1.2; the lower bound uses *exponential tilting* (change of measure) to make a the mean under the new measure, then CLT. [fill in: construct the tilted measure $d\mathbb{Q} \propto e^{\lambda x}\, d\mathbb{P}$] ∎

[mgf ↩] This thread is upgraded in §2 to "sub-Gaussian ≡ MGF dominated by $e^{\sigma^2\lambda^2/2}$".

The three deviation regimes (keep this picture): small deviations → normal; large deviations $\lambda_n = o(\sigma_n) \to \log\Pr \sim -\lambda_n^2/2$; very large deviations $\lambda_n = \Omega(\sigma_n) \to$ normal fails, Poisson takes over. Cf. Appendix A.

2 SUB-GAUSSIAN AND SUB-EXPONENTIAL

□□□□□ Turn ``what does the tail look like'' into a norm. Once classified, tail bounds come for free.

> In one line: rather than recompute MGFs each time, classify random variables by the *shape of their tail*. Sub-Gaussian = tail no heavier than a Gaussian's; sub-exponential = no heavier than an exponential's. Each family gets an Orlicz norm, and concentration becomes arithmetic on those norms.

The paradigm

Trigger. You want to handle a whole class of tails at once, and have the tail of sums and products follow *mechanically*.

Four equivalent characterizations (sub-Gaussian, write $\|X\|_{\psi_2}$): tail $\Pr[|X| > t] \le 2e^{-t^2/K^2}$ ⇔ moments $\|X\|_p \lesssim K\sqrt{p}$ ⇔ MGF $\mathbb{E}e^{\lambda X} \le e^{K^2\lambda^2/2}$ (after centering) ⇔ $\mathbb{E}e^{X^2/K^2} \le 2$.

> Three-line heuristic: (1) **a centered bounded variable is sub-Gaussian** (Hoeffding's lemma), so the entire ``bounded-difference'' family lands in range here; (2) sums of sub-Gaussians stay sub-Gaussian and the proxy variances *add*: $\|\sum X_i\|^2_{\psi_2} \lesssim \sum \|X_i\|^2_{\psi_2}$; (3) a product of two sub-Gaussians is sub-*exponential*, which is exactly why covariance / quadratic-form estimates drop to the sub-exponential family.

Cheat-sheet: the two tail families

Theorem	Setting	Tail bound / when to use
Hoeffding lemma	$\mathbb{E}X = 0$, $X \in [a, b]$	$\mathbb{E}e^{\lambda X} \le e^{\lambda^2(b-a)^2/8}$; bridge bounded → sub-Gaussian
Hoeffding ineq.	$X_i \in [a_i, b_i]$ independent	$\Pr[\lvert S - \mathbb{E}S\rvert \ge t] \le 2e^{-2t^2/\sum(b_i-a_i)^2}$
General Hoeffding	X_i sub-Gaussian	$\Pr[\lvert\sum a_i X_i\rvert \ge t] \le 2e^{-ct^2/\lVert a\rVert_2^2 K^2}$
Bernstein	X_i sub-exp., $\mathbb{E}X_i = 0$	$\Pr[\lvert S\rvert \ge t] \le 2\exp\big[-c\min(\frac{t^2}{\sum v_i^2}, \frac{t}{\max b_i})\big]$
Bennett	$\lvert X_i\rvert \le b$, variance σ^2	variance-aware; Gaussian for small t, Poisson for large t

Main results & proof skeletons

Definition 2.1 *(Sub-Gaussian norm).* $\|X\|_{\psi_2} = \inf\{K > 0 \ :\ \mathbb{E}\exp(X^2/K^2) \le 2\}$. Call X **sub-Gaussian** if this is finite. (Sub-exponential $\|X\|_{\psi_1}$ replaces X^2 by $|X|$.)

Proposition 2.2 *(Four equivalences).* For centered X the four characterizations above are pairwise equivalent, with constants differing only by absolute factors.

Proof. Skeleton: tail ⇒ moments (integrate the tail) ⇒ MGF (sum the series $\sum K^p p^{p/2} \lambda^p / p!$) ⇒ tail (Chernoff, §1) ⇒ Orlicz. [fill in: track the constants through the four steps; draw it as a cycle]

Lemma 2.3 *(Hoeffding's lemma).* $\mathbb{E}X = 0$, $X \in [a, b]$ a.s. ⇒ $\mathbb{E}e^{\lambda X} \le e^{\lambda^2(b-a)^2/8}$.

Proof. Skeleton: set $\psi(\lambda) = \log \mathbb{E}e^{\lambda X}$; compute $\psi(0) = \psi'(0) = 0$ and $\psi''(\lambda) = \mathrm{Var}_\lambda(X) \le (b-a)^2/4$ (the variance under the tilted measure is at most a quarter of the range squared); Taylor remainder. [fill in: prove $\psi'' \le (b-a)^2/4$ via the variance-range estimate]

Theorem 2.4 *(Hoeffding's inequality).* $X_i \in [a_i, b_i]$ independent, $S = \sum X_i$. Then $\Pr[|S - \mathbb{E}S| \ge t] \le 2\exp\big(-\frac{2t^2}{\sum_i (b_i-a_i)^2}\big)$.

Proof. [fill in: feed Lem 2.3 into the master bound of §1 + Lem 1.2, optimize λ]

Theorem 2.5 *(Bernstein's inequality).* X_i independent, centered, sub-exponential. Then the tail is Gaussian for small t and exponential for large t (see the cheat-sheet row).

Proof. Skeleton: the single-term sub-exponential MGF bound $\mathbb{E}e^{\lambda X_i} \le e^{v_i^2\lambda^2/2}$ holds only for $|\lambda| < 1/b_i$, so the choice of λ is constrained by $1/b$, producing the two-regime min. [fill in: do the constrained λ-optimization (the boundary between the two regimes)]

> Why it goes this way: a Gaussian tail comes from ``MGF dominated everywhere by $e^{c\lambda^2}$''; an exponential tail comes from ``dominated only for $|\lambda| < 1/b$''. The constrained λ-

optimization automatically yields Bernstein's $\min(t^2/v,\ t/b)$ split, which is the whole secret of being *variance-aware*.

Example 2.6 *(Johnson – Lindenstrauss, preview).* The squared length of a Gaussian random projection $\frac{1}{\sqrt{m}}Gx$ is a χ^2-type sum (sub-exponential), and Bernstein gives $\Pr[|\,\|\Pi x\|^2 - \|x\|^2\,| > \epsilon\|x\|^2] \le 2e^{-cm\epsilon^2}$. [fill in: add the union bound giving $m = O(\epsilon^{-2}\log N)$]

[mgf ↩] The sub-Gaussian thread is reused throughout §3 (McDiarmid) and §7 (matrix Bernstein) via [subgaussian ↩].

⇝ subgaussian
Check the constant: does Hoeffding's $2t^2/\sum(b-a)^2$ agree with Appendix A's A.1.18 (range ≤ 1 gives $e^{-2a^2/n}$)?

3 BOUNDED DIFFERENCES AND MARTINGALES

□□□□□ Upgrade ``sum of independents'' to ``any function that changes little when one coordinate moves''.

In one line: it need not be a sum. As long as $f(x_1,\dots,x_n)$ changes by a bounded amount when a single coordinate is altered, reveal its randomness *one coordinate at a time* (the Doob martingale); each step has bounded difference, feed it to Hoeffding's lemma from §2, and concentration appears.

The paradigm

Trigger. f is not a sum, but has **bounded differences**: altering coordinate i changes it by $|f - f'| \le c_i$.

Build the **Doob martingale** $M_k = \mathbb{E}[f \mid X_1,\dots,X_k]$, so $M_0 = \mathbb{E}f$, $M_n = f$, with increments $|M_k - M_{k-1}| \le c_k$. [subgaussian ↩]Apply Hoeffding's lemma per step and sum.

Three-line heuristic: (1) **any** function is its own Doob martingale (revealing one coordinate = one step); (2) concentration depends only on *how far each step can jump*, not on how complex f is; (3) the martingale constant ($\sum c_i^2$) is looser than the entropy method (§4) by a factor, but the proof is shortest and the assumptions weakest.

Cheat-sheet: the bounded-difference family

Theorem	Setting	Tail bound / when to use
Azuma–Hoeffding	increments $\lvert M_k - M_{k-1}\rvert \le c_k$	$\Pr[\lvert M_n - M_0\rvert \ge t] \le 2e^{-t^2/2\sum c_k^2}$
McDiarmid	f bounded diff. c_i, X_i indep.	$\Pr[\lvert f - \mathbb{E}f\rvert \ge t] \le 2e^{-2t^2/\sum c_i^2}$; workhorse of ML generalization
Self-bounding	f self-bounding	one-sided Bernstein type, variance-aware (cf. §4)
Freedman	martingale + predictable var. V	variance-aware martingale tail, tighter when V small

Main results & proof skeletons

Theorem 3.1 *(Azuma–Hoeffding).* (M_k) a martingale, $|M_k - M_{k-1}| \le c_k$ a.s. Then $\Pr[M_n - M_0 \ge t] \le \exp\big(-t^2/2\sum_k c_k^2\big)$.

Proof. Skeleton: for the increments $D_k = M_k - M_{k-1}$ we have $\mathbb{E}[D_k \mid \mathscr{F}_{k-1}] = 0$ and $|D_k| \le c_k$, so the conditional Hoeffding lemma gives $\mathbb{E}[e^{\lambda D_k} \mid \mathscr{F}_{k-1}] \le e^{\lambda^2 c_k^2/2}$; peel off layer by layer via the tower property to get $\mathbb{E}e^{\lambda(M_n - M_0)} \le e^{\lambda^2 \sum c_k^2/2}$; then Chernoff. [fill in: fill the conditional Hoeffding lemma and the tower-property unrolling]

Theorem 3.2 *(McDiarmid's bounded-differences inequality).* $X_1, \dots, X_n$ independent, f with bounded differences c_i. Then $\Pr[|f - \mathbb{E}f| \ge t] \le 2\exp\big(-2t^2/\sum_i c_i^2\big)$.

Proof. [fill in: feed the Doob martingale $M_k = \mathbb{E}[f \mid X_{1:k}]$ into Thm 3.1; independence keeps each increment's range $\le c_k$ (sup minus inf)]

Example 3.3 *(Concentration of the generalization gap).* $f = \sup_{h\in\mathscr{H}}\big(\mathbb{E}_{\mathscr{D}}\ell_h - \widehat{\mathbb{E}}\,\ell_h\big)$ has bounded differences $\le 1/n$ in each sample, so McDiarmid gives $\Pr[f - \mathbb{E}f \ge t] \le e^{-2nt^2}$. Thus the generalization gap concentrates uniformly around its expectation (= the Rademacher complexity). [fill in: verify the bounded-difference constant $1/n$]

> Why it goes this way: McDiarmid splits ``uniform convergence'' into two things, *concentration* (this section, f hugs $\mathbb{E}f$) and *the size of the expectation* ($\mathbb{E}f$ = Rademacher complexity, estimated by the chaining of §8). That is the source of the standard two-step recipe in statistical learning theory.

The truly tight bound comes from the *tensorization* of §4; this section is its "martingale approximation". [tensorize ↩]

⤳tensorize

Bounded differences can be relaxed: the asymmetric / variance-aware refinement (BLM Ch. 6) replaces $\sum c_i^2$ by the true variance, but needs the entropy method. Room to improve here.

4 THE VARIANCE AND ENTROPY METHODS

□□□□□ Tensorization: variance and entropy are subadditive over product measures, so high-dim = one-dim subadditivity + a single-coordinate estimate.

> In one line: both variance (Efron – Stein) and entropy (log-Sobolev / Herbst) are *subadditive* over product measures. **Tensorize** a high-dimensional concentration into n single-coordinate contributions, each needing only a one-dimensional estimate. This route gives the tightest, *variance-aware, dimension-free* bounds, and is the analytic engine behind the geometry of §5 and Talagrand (§6).

The paradigm

Trigger. You want the tightest *variance-aware* bound, or to prove a function sub-Gaussian without a "sum / martingale" structure.

Herbst's argument: if the entropy obeys $\mathrm{Ent}[e^{\lambda f}] \le \frac{\lambda^2\sigma^2}{2}\,\mathbb{E}[e^{\lambda f}]$ for all λ, then $f - \mathbb{E}f$ is σ-sub-Gaussian. Combine with the **tensorization** of entropy $\mathrm{Ent}[g] \le \sum_i \mathbb{E}\,\mathrm{Ent}_i[g]$ to reduce the problem to a single coordinate.

Three-line heuristic: (1) **Efron–Stein** is the tensorization of variance, giving the *free* (no i.i.d. needed) bound $\operatorname{Var} f \le \frac{1}{2}\sum_i \mathbb{E}(f - f^{(i)})^2$; (2) replace Var by Ent and ``square'' by ``exponential'': tensorization + a one-dimensional log-Sobolev = exponential concentration (Herbst); (3) the Gaussian measure satisfies a *dimension-free* log-Sobolev inequality (Gross), the algebraic source of the Gaussian concentration in §5.

Cheat-sheet: variance and entropy

Theorem	Setting	Gives / when to use
Efron–Stein	product measure, $f^{(i)}$ = resample coord. i	$\operatorname{Var} f \le \frac{1}{2}\sum_i \mathbb{E}(f - f^{(i)})^2$; the "bounded difference" for variance
Bounded diff. via ES	bounded diff. c_i	$\operatorname{Var} f \le \frac{1}{4}\sum c_i^2$; variance from the martingale family
Herbst argument	$\operatorname{Ent}[e^{\lambda f}] \le \frac{\lambda^2\sigma^2}{2}\mathbb{E}e^{\lambda f}$	$\Rightarrow$ f is σ-sub-Gaussian
Gaussian LSI	$\gamma = N(0, I_n)$	$\operatorname{Ent}[g^2] \le 2\mathbb{E}\|\nabla g\|^2$; dimension-free
Modified LSI	bounded / Bernoulli coords	tensorize + single-coord LSI $\Rightarrow$ concentration

Main results & proof skeletons

Theorem 4.1 *(Efron–Stein inequality).* $X_1, \dots, X_n$ independent, X_i' an i.i.d. copy, $f^{(i)}$ replacing coordinate i by X_i'. Then $\operatorname{Var} f \le \frac{1}{2}\sum_{i=1}^{n} \mathbb{E}\big[(f - f^{(i)})^2\big]$.

Proof. Skeleton: use the orthogonal martingale-difference decomposition $\operatorname{Var} f = \sum_i \mathbb{E}[\Delta_i^2]$ with $\Delta_i = \mathbb{E}[f \mid X_{1:i}] - \mathbb{E}[f \mid X_{1:i-1}]$; then Jensen to replace Δ_i by the symmetrized $(f - f^{(i)})$. [fill in: fill the orthogonal decomposition + symmetrization, two steps] ∎

Theorem 4.2 *(Herbst's argument).* If $\operatorname{Ent}[e^{\lambda f}] \le \frac{\lambda^2\sigma^2}{2}\mathbb{E}[e^{\lambda f}]$ for all λ, then $\mathbb{E}[e^{\lambda(f - \mathbb{E}f)}] \le e^{\lambda^2\sigma^2/2}$, so f is σ-sub-Gaussian.

Proof. Skeleton: let $H(\lambda) = \frac{1}{\lambda}\log \mathbb{E}e^{\lambda f}$; the entropy inequality is exactly the rewriting $H'(\lambda) \le \sigma^2/2$; integrate to get $H(\lambda) - H(0^+) \le \lambda\sigma^2/2$, with $H(0^+) = \mathbb{E}f$. [fill in: fill the differential identity $\operatorname{Ent}[e^{\lambda f}] = \lambda^2 \mathbb{E}e^{\lambda f} H'(\lambda)$] ∎

Theorem 4.3 *(Gaussian log-Sobolev, Gross).* For $\gamma = N(0, I_n)$ and smooth g, $\operatorname{Ent}_\gamma[g^2] \le 2\,\mathbb{E}_\gamma\|\nabla g\|^2$.

Proof. [fill in: fill: the one-dimensional LSI (Ornstein–Uhlenbeck semigroup, or CLT limit from two-point Bernoulli) + tensorization to n dimensions] ∎

Why it goes this way: Gaussian LSI + Herbst ⇒ ``Gaussian concentration $e^{-t^2/2L^2}$ for an L-Lipschitz function, dimension-free''. This is the algebraic path that *computes* the geometric statement of §5: curvature (the LSI constant) turns directly into the tail rate.

[tensorize ↩] Tensorization reappears in §6 in Marton's transportation form.

Four faces: concentration ⇔ isoperimetry ⇔ log-Sobolev ⇔ transportation (Marton). Herbst is the "LSI ⇒ concentration" face; the other three are to be filled, see §5 and §9 open threads.

5 GEOMETRY AND ISOPERIMETRY

□□□□□ Where "concentration of measure" gets its name: on a metric measure space, a Lipschitz function is almost constant.

> In one line: a viewpoint with no MGF at all. On a metric measure space, every 1-Lipschitz function **hugs its median**; this is equivalent to an **isoperimetric** phenomenon, a set of measure $\geq \frac{1}{2}$, blown up by t, swallows $1 - e^{-ct^2}$ of the measure. High-dimensional spheres and Gaussian space behave exactly so; this is the geometric root of "a random vector's length is almost determined" and "a random projection preserves distance".

The paradigm

Trigger. f is an L-Lipschitz function on a space with **concentration function** $\alpha(t)$.

Then $\Pr[|f - \mathrm{Med}\, f| > t] \leq 2\,\alpha(t/L)$. Three pillar spaces:

$$\begin{aligned} \text{sphere } S^{n-1} &: \alpha(t) \leq e^{-(n-1)t^2/2}; \\ \text{Gaussian } \gamma_n &: \alpha(t) \leq e^{-t^2/2} \quad (\textbf{dimension-free}); \\ \text{Hamming cube} &: \alpha(t) \leq e^{-2t^2/n}. \end{aligned}$$

> Three-line heuristic: (1) concentration ≡ isoperimetry: the tail bound is the complement of "the t-blow-up of a half-measure set"; (2) median and mean are interchangeable (differ by $O(L)$, same order); (3) the Gaussian case being *dimension-free* is the key miracle, it makes infinite-/high-dimensional processes (§8) possible.

Cheat-sheet: geometric concentration

Theorem	Space	Statement / when to use
Lévy's lemma	sphere S^{n-1}	1-Lip f: $\Pr[\lvert f - \mathrm{Med}\, f\rvert > t] \leq 2e^{-(n-1)t^2/2}$
Gaussian conc.	$\gamma_n = N(0, I_n)$	L-Lip f: $\Pr[\lvert f - \mathbb{E} f\rvert > t] \leq 2e^{-t^2/2L^2}$, *dimension-free*
Borell–TIS	Gaussian process	$\sup_t X_t$ concentrates around its supremum, variance $= \sup_t \mathrm{Var}\, X_t$
Gaussian isoper.	γ_n	half-spaces are extremal; exact blow-up in Φ^{-1} form

Main results & proof skeletons

> **Theorem 5.1** *(Lévy's lemma / spherical concentration).* $f : S^{n-1} \to \mathbb{R}$ is 1-Lipschitz. Then $\Pr[|f - \mathrm{Med}\, f| > t] \leq 2e^{-(n-1)t^2/2}$.

Proof. Skeleton: spherical **isoperimetry** (Lévy–Schmidt: geodesic balls are extremal) ⇒ for a half-measure set A, the t-neighborhood satisfies $\sigma(A_t) \geq 1 - e^{-(n-1)t^2/2}$; take $A = \{f \leq \mathrm{Med}\, f\}$, and Lipschitz pushes $\{f > \mathrm{Med}\, f + t\}$ outside A_t. [fill in: fill the standard isoperimetry ⇒ concentration translation; cite spherical isoperimetry (do not reprove)] ∎

Theorem 5.2 *(Gaussian concentration / Borell–TIS).* $f : \mathbb{R}^n \to \mathbb{R}$ is L-Lipschitz, $X \sim N(0, I_n)$. Then $\Pr[|f(X) - \mathbb{E}f| > t] \leq 2e^{-t^2/2L^2}$, with constants independent of n.

Proof. Either path: *(a)* Gaussian log-Sobolev (Thm 4.3) + Herbst (Thm 4.2), with smooth approximation of f; or *(b)* Gaussian isoperimetry (half-space extremality) gives the blow-up directly. [fill in: take (a): fill the Lipschitz-gradient equivalence $\|\nabla f\| \leq L$ a.e. + smoothing]

Example 5.3 *(A random vector's length is almost determined).* $x \mapsto \|x\|$ is 1-Lipschitz ⇒ for $X \sim N(0, I_n)$, $\|X\|$ concentrates near $\sqrt{n}$ with fluctuation $O(1)$ (independent of n). This is the one-line proof of the "thin-shell" phenomenon. [fill in: fill $\mathbb{E}\|X\| = \sqrt{n}(1 - o(1))$ and substitute]

> Why it goes this way: the MGF method (§1) answers ``sums''; the geometric method answers ``*any* Lipschitz function'', dimension-free in the Gaussian case, at the cost of needing a ``hard'' input like isoperimetry / log-Sobolev. The two paths *meet* at Gaussian concentration: the LSI of §4 is precisely the algebraic proof of Borell–TIS.

Borell–TIS (concentration of the supremum of a Gaussian process) is the "concentration half" of the chaining in §8, chaining controls the *expectation*, this theorem the *fluctuation*. [borelltis ↩]

borelltis Median vs mean: they differ by $O(L)$, but exact isoperimetry is most natural with the median. When must they be distinguished? (heavy tails, asymmetric f.)

6 TALAGRAND'S CONVEX DISTANCE

□□□□□ A product space has no geometry, only convexity. Talagrand's convex distance adapts to ``how many bad coordinates''.

> In one line: on a pure *product* probability space (no sphere-like geometry), Talagrand's **convex distance** yields dimension-free, very strong concentration for convex Lipschitz functions. Its edge over bounded differences (§3): it does not look at the ``largest single-coordinate jump'' but at ``how *many* coordinates you must change to escape a set'', adapting to sparse dependence.

The paradigm

Trigger. f on a product space is **convex** Lipschitz, or escaping an event A requires changing many coordinates.

Define the **convex distance** $d_T(x, A) = \sup_{\|\alpha\|_2 \leq 1} \inf_{y \in A} \sum_{i : x_i \neq y_i} \alpha_i$ (weighted Hamming). Talagrand: $\mathbb{E}\big[e^{d_T(X,A)^2/4}\big] \leq 1/\Pr[A]$, hence

$$\Pr[A] \cdot \Pr[d_T(X, A) \geq t] \leq e^{-t^2/4}.$$

> Three-line heuristic: (1) d_T is a min-max of *weighted* Hamming, automatically finding the ``cheapest escape direction''; (2) once $\Pr[A] \geq \frac{1}{2}$, the convex distance from A concentrates sub-Gaussianly; (3) for a *convex* L-Lipschitz f, take $A = \{f \leq \mathrm{Med}\, f\}$ to get $\Pr[|f - \mathrm{Med}\, f| > t] \leq 4e^{-t^2/4L^2}$, dimension-free.

Cheat-sheet: convex distance and its corollaries

Result	Setting	Statement / when to use
Convex-distance ineq.	product space, any A	$\Pr[A]\Pr[d_T \ge t] \le e^{-t^2/4}$; the master switch
Convex Lipschitz	f convex, L-Lip on $[0,1]^n$	$\Pr[\|f-\mathrm{Med}\,f\| > t] \le 4e^{-t^2/4L^2}$
LIS / combinatorial opt.	f = longest incr. subseq., etc.	$\sqrt{}$-scale fluctuation, beats bounded diff.'s n-scale
Largest singular value	random matrix	$\sigma_{\max}$ concentrates around its median

Main results & proof skeletons

Theorem 6.1 *(Talagrand's convex-distance inequality).* $X = (X_1,\dots,X_n)$ a product measure, A measurable. Then $\mathbb{E}\big[e^{d_T(X,A)^2/4}\big] \le 1/\Pr[A]$.

Proof. Skeleton (induction on n, the hardest step in the book): *(i)* write d_T via convex duality (the distance to the convex hull $U_A(x) = \mathrm{conv}\{\mathbf{1}[x_i \ne y_i] : y \in A\}$); *(ii)* condition on the last coordinate and apply the inductive hypothesis to the two cases "in A / projection in A"; *(iii)* a one-dimensional optimization $\inf_{0\le r\le 1} r^{\cdots}\cdots$ merges the two cases. [fill in: fill the three-step induction; the one-dim optimization lemma $\inf_r e^{(1-r)^2/4}(\cdot)^r$ is the crux] ∎

Corollary 6.2 *(Concentration of convex Lipschitz functions).* $f : [0,1]^n \to \mathbb{R}$ convex and L-Lipschitz (Euclidean), X a product measure. Then $\Pr[|f - \mathrm{Med}\,f| > t] \le 4e^{-t^2/4L^2}$.

Proof. [fill in: use convexity to control $f(x) - \mathrm{Med}\,f$ by $L \cdot d_T(x, \{f \le \mathrm{Med}\,f\})$ (convex ⇒ supporting hyperplane), then apply Thm 6.1] ∎

Example 6.3 *(Longest increasing subsequence (LIS)).* For a random permutation, $L_n \sim 2\sqrt{n}$; changing one value moves it by at most 1, but "certifying $L_n \ge k$ needs only k positions", so the convex distance gives fluctuation $O(n^{1/4})$, far beating bounded differences' $O(\sqrt{n})$. [fill in: fill the certifiable / self-certifying structure ⇒ d_T bound]

Why it goes this way: bounded differences only knows ``each coordinate has $c_i$''; Talagrand knows ``how *many* coordinates must conspire to change the conclusion''. When dependence is sparse (few coordinates determine f), convex distance replaces $\sqrt{\sum c_i^2}$ by $\sqrt{\#\text{critical coords}}$, the source of the $n \to \sqrt{n} \to n^{1/4}$ chain of improvements in combinatorial optimization.

[tensorize ↩] Convex distance can also be seen as the scalar precursor of the spectral concentration of §7.

Another route: Marton's transportation-cost ($W_1 \le \sqrt{2\,\mathrm{KL}}$) also gives convex-distance concentration, tying it back to the entropy tensorization of §4. Worth listing side by side.

7 MATRIX CONCENTRATION

□□□□□ Lift scalar Chernoff/Bernstein to the spectral norm of a random matrix. The obstacle is $e^{A+B} \ne e^A e^B$.

In one line: to bound the largest eigenvalue of a sum of independent random matrices, copying §1 stalls because *matrix exponentials do not multiply*. The rescue is **Lieb's concavity** + the trace exponential: use $\mathbb{E}\,\operatorname{tr}\exp(\cdot)$ as a surrogate; it is *subadditive*, so the scalar proof transfers almost verbatim, with the dimension entering only at the end as $\log d$.

The paradigm

Trigger. $\sum_i X_i$ is a sum of independent random Hermitian matrices and you want a tail for $\|\sum_i X_i\|$ (spectral norm).

Matrix Laplace transform: for $\theta > 0$,

$$\Pr\left[\lambda_{\max}(\textstyle\sum_i X_i) \ge t\right] \le e^{-\theta t}\,\mathbb{E}\,\operatorname{tr}\exp\Big(\textstyle\sum_i \log \mathbb{E}\, e^{\theta X_i}\Big),$$

where the $\operatorname{tr}\exp(\sum\cdots)$ comes from Lieb's theorem ($A \mapsto \operatorname{tr}\exp(H + \log A)$ is concave) + Jensen.

Three-line heuristic: (1) turn ``$\lambda_{\max} \ge t$'' into $\operatorname{tr} e^{\theta \sum X_i} \ge e^{\theta t}$ then Markov; (2) **Lieb** lets us push $\mathbb{E}$ inside $\operatorname{tr}\exp$ and makes the matrix cumulants *subadditive* (replacing the scalar ``MGFs multiply''); (3) the dimension d enters only through $\operatorname{tr} \le d \cdot \lambda_{\max}$, so it is nearly dimension-free at *low intrinsic dimension*.

Cheat-sheet: matrix tail bounds

Theorem	Setting	Tail bound / when to use
Matrix Laplace	independent Hermitian X_i	the display above; mother of all matrix tails
Matrix Chernoff	$0 \preceq X_i \preceq R\,I$	$\lambda_{\min/\max}(\sum X_i)$ around $\mu_{\min/\max}$; PSD sums
Matrix Bernstein	$\mathbb{E}X_i = 0$, $\|X_i\| \le L$, $\|\sum \mathbb{E}X_i^2\| = v$	$\Pr[\|\sum X_i\| \ge t] \le 2d\, e^{-t^2/2(v+Lt/3)}$
Intrinsic-dim.	as above, intdim $= v/\|\cdot\|$	$d \to$ intrinsic dim, drops the ambient-dimension log

Main results & proof skeletons

Theorem 7.1 *(Matrix Laplace transform method, Tropp).* For independent random Hermitian X_i and $\theta > 0$, the matrix Laplace inequality above holds.

Proof. Skeleton: *(i)* $\Pr[\lambda_{\max} \ge t] \le e^{-\theta t}\mathbb{E}\,\operatorname{tr} e^{\theta\sum X_i}$ (Markov + $\lambda_{\max} \le \operatorname{tr}$ on PSD); *(ii)* **Lieb's theorem** $\Rightarrow$ $A \mapsto \operatorname{tr}\exp(H + \log A)$ concave $\Rightarrow$ Jensen pushes $\mathbb{E}$ inside; *(iii)* peel off term by term to get the subadditive cumulant $\operatorname{tr}\exp(\sum_i \log \mathbb{E}e^{\theta X_i})$. [fill in: fill the Lieb $\Rightarrow$ subadditivity induction; cite Lieb (do not reprove)] ∎

Theorem 7.2 *(Matrix Bernstein).* X_i independent, $\mathbb{E}X_i = 0$, $\|X_i\| \le L$, variance proxy $v = \|\sum_i \mathbb{E}X_i^2\|$. Then $\Pr[\|\sum_i X_i\| \ge t] \le 2d\exp\!\big(\frac{-t^2/2}{v+Lt/3}\big)$.

Proof. [fill in: feed the single-term matrix cgf bound $\log \mathbb{E}e^{\theta X_i} \preceq \frac{\theta^2/2}{1-L\theta/3}\mathbb{E}X_i^2$ into Thm 7.1, optimize θ] ∎

Example 7.3 *(Covariance estimation).* For the sample covariance $\widehat{\Sigma}$ of n independent samples, write $\widehat{\Sigma} - \Sigma = \frac{1}{n}\sum_i X_i$ (centered outer products); matrix Bernstein gives $\|\widehat{\Sigma} - \Sigma\| \lesssim \sqrt{\frac{\|\Sigma\| r \log d}{n}}$ (r = intrinsic dimension). [fill in: fill the L, v estimates of the outer products and the sample-size threshold]

> Why it goes this way: the entire structure of scalar Bernstein (the variance term v + the uniform bound L, the two-regime $\min$) survives untouched, with only an extra d (or intrinsic-dim) prefactor, coming from turning ``one tail'' into ``the tail of d eigenvalues''. This machine replaces the ``union bound over an ϵ-net'' common in ML proofs, and is both tighter and less work.

This thread feeds directly into sketching and leverage-score sampling. [subgaussian ↩]

matrixbern

Martingale version: matrix Freedman (Tropp 2011) extends this to matrix martingales, pairing with §3. Needed for the spectral concentration of online / adaptive sampling.

8 SUPREMA AND CHAINING

□□□□□ The exit. When the target is $\sup_t X_t$ (an empirical process), single-point Chernoff + a union bound is too loose.

> In one line: the first seven sections control *one* random quantity. But ML/statistics really wants $\sup_{t\in T} X_t$ (empirical processes, the sup in a generalization bound). **Chaining** uses multi-scale nets to control the supremum by an integral of metric entropy, reducing ``infinitely many tails'' to ``the geometry of covering numbers''. This is the bridge to empirical processes.

The paradigm

Trigger. You want $\mathbb{E}\sup_{t\in T} X_t$, where $\{X_t\}$ is a (sub-)Gaussian process with natural metric $d(s,t) = \|X_s - X_t\|_{\psi_2}$.

Chaining: along a sequence of refining ε-nets $T_0 \subset T_1 \subset \cdots$, telescope $X_t - X_{t_0}$ and sum scale by scale. **Dudley**:

$$\mathbb{E}\sup_{t\in T} X_t \;\le\; C\int_0^\infty \sqrt{\log N(T,d,\varepsilon)}\,d\varepsilon.$$

> Three-line heuristic: (1) a union bound = a single scale, chaining = *all scales* summed, hence tighter; (2) the currency is **metric entropy** $\log N(T,d,\varepsilon)$, turning a probability problem into covering geometry; (3) Dudley can be loose; the **generic chaining** quantity $\gamma_2(T,d)$ is the *tight* one (Talagrand's majorizing measures).

Cheat-sheet: from union bound to majorizing measures

Tool	Gives	When to use / tight?
Union bound	$\sqrt{2\log\lvert T\rvert}$ (finite T)	opening move; too loose for large or continuous T
Dudley integral	$\int\sqrt{\log N(\varepsilon)}\,d\varepsilon$	standard upper bound for continuous index; often enough
Sudakov lower bd.	$\sup_\varepsilon \varepsilon\sqrt{\log N(\varepsilon)}$	pairs with Dudley to pin the order
Generic chaining	$\gamma_2(T,d)$	matching upper *and* lower (Gaussian processes, Talagrand)

Main results & proof skeletons

> **Theorem 8.1** *(Dudley's inequality).* $\{X_t\}_{t\in T}$ a centered process, sub-Gaussian w.r.t. d. Then $\mathbb{E}\sup_t X_t \le C\int_0^\infty \sqrt{\log N(T,d,\varepsilon)}\,d\varepsilon$.

Proof. Skeleton: take nets T_k at scale $\varepsilon_k = 2^{-k}$ with nearest-point maps $\pi_k(t)$; telescope $X_t - X_{\pi_0(t)} = \sum_{k\ge1}(X_{\pi_k(t)} - X_{\pi_{k-1}(t)})$; each link has $|T_k||T_{k-1}|$ sub-Gaussian increments, combine the per-scale union bound with the increment scale $\lesssim 2^{-k}$ and sum. [fill in: fill the per-scale union bound $\sqrt{\log|T_k|}$ and the geometric-series sum ⇒ the integral]

> **Theorem 8.2** *(Generic chaining / majorizing measures).* For a Gaussian process, $\mathbb{E}\sup_t X_t \asymp \gamma_2(T,d)$, where $\gamma_2(T,d) = \inf_{(T_k)}\sup_t\sum_{k\ge0} 2^{k/2} d(t,T_k)$ with $|T_k|\le 2^{2^k}$.

Proof. [fill in: upper bound: weighted chaining (weights adapt to t, beating Dudley's uniform scales); lower bound: Talagrand's partition tree / iterated Sudakov. Cite the majorizing-measures theorem (do not reprove)]

> **Example 8.3** *(Operator norm via ε-net + chaining).* $\|A\| = \sup_{u,v\in S^{n-1}} u^\top Av$: discretize on a $1/2$-net of the sphere ($\le 9^n$ points) + each point sub-Gaussian (§2) ⇒ $\mathbb{E}\|A\| \lesssim \sqrt{n}$. Chaining "automates" the net argument to a continuous index. [fill in: fill the standard $1/2$-net lemma lifting sup to $2\max_{\text{net}}$]

> Why it goes this way: the Borell–TIS of §5 ([borelltis ↩]) controls the *fluctuation* of $\sup_t X_t$, this section its *expectation*, together they are the full concentration of an empirical process. This closes the loop with the ``expectation = Rademacher complexity'' half of the generalization bound in §3.

Chaining → Rademacher complexity → uniform laws of large numbers. ⇝ chaining

OPEN THREADS · WHERE THIS GOES NEXT

□□□□□

> These notes are an **arsenal**, not a museum. The cross-section open questions below are the seeds for the next pass: a unifying perspective, the missing lower bounds, and the exit toward empirical processes.

Open threads (cross-section)

- **Four faces**: concentration ⇔ isoperimetry ⇔ log-Sobolev ⇔ transportation (Marton). Drawing the explicit commuting diagram of these equivalences is the "main theorem" this notebook most wants to add.
- **When does dimension enter**? Gaussian concentration is dimension-free, matrix concentration carries $\log d$, chaining carries metric entropy. A "sources of dimension dependence" table separates "truly dimension-free" from "hidden in the constants".
- **Tightness**: is every upper bound paired with a lower bound? Cramér (§1) and Sudakov (§8) are; the lower bounds for McDiarmid and matrix Bernstein (worst f / worst spectrum) are to be filled.
- **The exit**: after §8, open a dedicated *empirical-process* notebook (VC / Rademacher / uniform laws / Donsker).

Once this diagram is in, §4–§6 unify into one engine.

> Closing line: the whole subject is seven variations on one move, *exponentiate the deviation, then crush it with structure*. What the structure is decides which engine you reach for. Recognizing the structure beats memorizing the formula.